SMART CONTRACT SECURITY REVIEW

Mezo Auth Dev
Point-wise Findings with Risk Ratings

Scope and Rating Note

Preliminary review of the listed VEMEZO NFT, marketplace administration, treasury/payment components, and x402/Permit2 integration on Mezo Testnet (Chain ID 31611). Risk ratings are initial priority estimates based on component roles and potential impact; they are not proof of an exploitable vulnerability. Contract-level conclusions require verified deployed code and targeted tests.

Contract Address Register

Contract / Component	Exact Address	Stated Role
VEMEZO NFT	0xaCE816CA2bcc9b12C59799dcC5A959Fb9b98111b	NFT creation and marketplace asset
MarketplaceAdmin	0x1D462522A9f8fbea6c10122dce01E09432A64dBF	Marketplace administration and fee controls
MEZO Treasury	0x2Db34b3aBb62Ffcfe7F6927Cfa4F52C16D649694	Treasury destination
Governance	0x82531Fc1b2eB804D7484b1346fb0FE503F81c75a	Governance contract; exact responsibilities and permissions require confirmation
Mezo Caller/Token	0x7B7c000000000000000000000000000000000001	Pay-per-use integration component
MUSD Token / pay-per-use	0xD88b46ef8444dAA8aa493d714b1124DE75CCa067	Token configured for pay-per-use flow
MUSD Platform / payments	0x21f7C9fdA5ED418AF3a1C7593dec52142350A0F7	Payment platform and accounting path
Permit2	0x000000000022D473030F116dDEE9F6B43aC78BA3	Token-approval integration dependency
x402Permit2Proxy	0x8dea1b08dc2e1D9b556450f736F19968F367A98d	Custom proxy in x402/Permit2 flow
Settlement proxy	0x402085c248EeA27D92E8b30b2C58ed07f9E20001	x402 settlement integration
MUSD x402	0x118917a40FAF1CD7a13dB0Ef56C86De7973Ac503	MUSD token/component associated with x402
Oracle Data Feeds	0x2880aB155794e7179c9eE2e38200202908C17B43	External data-feed integration

Findings

F-01 - Treasury destination requires confirmation [MEDIUM RISK]

Finding: The listed MEZO Treasury address is identified as the destination for treasury funds. The supplied information does not confirm that this address matches the official deployment and operational configuration. **No major vulnerability is established by the address.**

F-02 - x402 settlement authorization and call path [MEDIUM RISK]

Finding: The payment flow includes Permit2, a custom x402Permit2Proxy, and a separate settlement proxy. The authorization checks, configured spender, and sequence of calls determine which component can initiate or complete a payment. This is an integration review concern, not a confirmed exploit. **No major vulnerability is established by the address.**

F-03 - Marketplace administrative controls [MEDIUM RISK]

Finding: MarketplaceAdmin is associated with marketplace administration and fee settings. Its privileged roles and access checks determine who can update configuration or change fees. If those controls are too broad or not properly enforced, unauthorized changes may be possible. **No major vulnerability is established by the address.**

SMART CONTRACT SECURITY REVIEW — FINDINGS

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F-04 - Payment accounting, withdrawals, and refunds [MEDIUM RISK]

Finding: The MUSD Platform is part of the payment flow. Its accounting, withdrawal, and refund behavior are relevant to balance accuracy and fund handling. **No major vulnerability is established by the address.**

F-05 - Token address consistency across flows [MEDIUM RISK]

Finding: Separate MUSD addresses are listed for pay-per-use and x402. The intended token for each flow should be clear and consistent with the deployment configuration. **No major vulnerability is established by the address.**

F-06 - Pay-per-use caller and token role separation [MEDIUM RISK]

Finding: The Mezo Caller/Token and MUSD pay-per-use components are listed separately. Their respective caller, executor, and token roles are not fully described in the supplied information. **No major vulnerability is established by the address.**

F-07 - VEMEZO NFT minting and marketplace permissions [MEDIUM RISK]

Finding: Relevant review areas include mint authorization, token-ID uniqueness, ownership changes, and marketplace approvals. No specific NFT exploit is identified in the supplied information. **No major vulnerability is established by the address.**

F-08 - MUSD/MEZO token interface and accounting assumptions [LOW RISK]

Finding: The listed MUSD and MEZO token integrations should be checked for consistent decimal normalization, safe transfer/transferFrom return-value handling, and allowance updates. Incorrect assumptions in these ERC-20 behaviors can cause payment amounts or recorded balances to diverge. **No major vulnerability is established by the address.**

F-09 - Governance privileged execution controls [LOW RISK]

Finding: The Governance address is listed, but its control model is not specified. The key technical boundary is whether proposal execution or administrative roles can change treasury, marketplace, or upgrade settings; proposal thresholds, quorum, executor permissions, and any timelock determine that scope. **No major vulnerability is established by the address.**

F-10 - Pyth price-feed validation and update handling [LOW RISK]

Finding: The listed address is the Pyth oracle proxy on Mezo Testnet. Consumers should bind each asset to the correct Pyth feed ID and validate publishTime freshness, confidence interval, and exponent/decimal normalization before using a price. Pull updates should also account for the required update fee. **No major vulnerability is established by the address.**

Risk Summary

Medium Risk: F-01, F-02, F-03, F-04, F-05, F-06, F-07
Low Risk: F-08, F-09, F-10

Overall Result

This preliminary review groups the listed findings as Medium or Low Risk. The ratings reflect review priority based on the component descriptions provided, not confirmed vulnerabilities. Final conclusions depend on verified deployed code, configuration, and targeted testing.

References

• Mezo Testnet contracts reference: https://testnet.mezo.org/docs/users/resources/contracts-reference
• Mezo MUSD payments and x402 documentation: https://mezo.org/docs/fr/docs/developers/getting-started/musd-payments-x402/
• Mezo Pyth oracle documentation: https://testnet.mezo.org/docs/developers/getting-started/integrations-and-partners/
• Mezo Testnet Explorer: https://explorer.test.mezo.org/